



PLANT DISEASES — FULL MIND MAP

Biology Ch. 9.vii · Diseases & Treatment · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Plant Pathogens, Causal Factors & Disease Spread
2. Important Crop–Disease Matching
3. Zinc Deficiency Disorders in Plants
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5. Major Rice Diseases
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7. Potato Blackheart — Oxygen Deficiency
8. Okra Yellow Vein Mosaic
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1. Plant Pathogens, Causal Factors & Disease Spread

- Viruses living cells ke andar replicate karte hain aur source ke according bacteria, fungi aur plants — teeno ko infect kar sakte hain. [Q1]
- Plant-disease causal matching: Citrus canker → *Xanthomonas* bacteria; sugarcane red rot → *Colletotrichum falcatum* fungus; potato Blackheart/Krishnakant → oxygen deficiency. Wheat ear-cockle/sehun ko source nematode se link karta hai. [Q2]
- Plant diseases ke dissemination factors mein wind, rain, animals, soil, contaminated tools, infected seed stock, pollen, dust storm, irrigation water, birds aur humans shamil hain; Q17 mein Birds + Dust + Rain + Wind sab correct hain. [Q17]



EXAM TRAP — Q2 — Disputed Matching

- Source kehta hai question mein wheat ka 'Sahu' disease diya hai jo 'sehun' hona chahiye; ear cockle/sehun nematode se hota hai, insect se nahi. Isliye diye gaye options mein perfect match nahi banta aur answer '*' disputed rakha gaya hai. [Q2]

2. Important Crop–Disease Matching

- Powdery mildew source ke according wheat, grape, barley, onion, apple aur pea se related hai — sugarcane se nahi. Wheat–Phalaris minor, Potato–Late blight aur Bajra–Ergot pairs correct diye gaye hain. [Q3]
- Green hair disease Bajra mein hota hai, rice mein nahi; bajra ke hairs twist hokar broom-like appearance dete hain aur plant dwarf reh sakta hai. Bajra–Ergot, Pea–Powdery mildew, Gram–Wilt correct pairs hain. [Q4]
- Wilt ko source gram/chickpea, sugarcane, toor etc. se link karta hai, paddy se nahi. Mustard–White rust, Bajra–Smut aur Groundnut–Tikka correct pairs diye gaye hain. [Q5]

3. Zinc Deficiency Disorders in Plants

- Maize mein Zinc (Zn) deficiency se chlorosis hota hai; leaves underdeveloped/dwarf ho sakte hain aur leaf tip white, mottled ya yellow ho sakta hai. [Q6, Q7]
- White Bud disease bhi Zn deficiency se linked hai. Source Zn-deficiency names mein rice ka Khaira, maize ka White Bud aur citrus/cotton ka mottle leaf (little leaf)/frenching deta hai. [Q8]

4. Sugarcane, Groundnut & Mustard Problems

- Pyrilla sugarcane ka harmful leafhopper pest hai jo leaf/plant sap suck karta hai; infestation se yellow patches, leaf drying aur crop growth disruption ho sakta hai. [Q9]
- Groundnut ki major/deadly disease Tikka hai; source ise early aur late leaf spots ke roop mein describe karta hai aur *Cercospora personata* / *C. arachidicola* fungi se link karta hai. [Q10]

- Source Q11 white rust ko mustard ki prominent fungal disease batata hai aur Albugo candida se link karta hai; stem, leaves aur inflorescence par white lesions/patches aa sakte hain. [Q11]

5. Major Rice Diseases

- Rice blast ek fungal disease hai; source causal fungus ko Pyricularia oryzae / Magnaporthe oryzae (M. grisea) batata hai. Ye above-ground rice parts ko affect kar sakta hai aur destructive crop loss cause karta hai. [Q12]
- False smut paddy ka disease hai, barley ka nahi. Source matching mein Wheat–Rust, Potato–Early blight aur Paddy–Khaira ko correct batata hai. [Q13]
- Rice Tungro do virus groups — RTSV (Rice Tungro Spherical Virus) aur RTBV (Rice Tungro Bacilliform Virus) — se linked hai; dono green leafhoppers ke through transmit hote hain. [Q16]

6. Major Wheat Diseases

- Wheat ke do common fungal diseases: Black rust → Puccinia graminis tritici; Loose smut → Ustilago nuda tritici. [Q14]
- Karnal Bunt wheat ki fungal disease hai; source causal smut fungus Tilletia indica batata hai. Iska first report 1931 mein Karnal, Haryana ke paas wheat se diya gaya hai. [Q15]

7. Potato Blackheart — Oxygen Deficiency

- Potato Blackheart ek abiotic storage disorder hai jo low oxygen availability se hota hai; tuber centre black ho sakta hai aur tissue inside-out die karke jet black ho jaata hai. [Q18]

8. Okra Yellow Vein Mosaic

- Okra/Bhindi ka Yellow Vein Mosaic ek viral disease hai; Whitefly iska vector/factor hai jo virus ko ek plant se doosre plant tak spread karta hai. [Q19, Q20]

9. Mango Spongy Tissue Disorder

- Spongy tissue Alphonso mango ka ripening disorder hai: fruit bahar se normal lag sakta hai lekin andar soft-centre/white corky tissue ya internal breakdown hota hai. Source ise South India se peculiar aur export quality ko hamper karne wala disorder batata hai. [Q21]